Case 03 · Research and POC: Exam Paper Time-Lock

Following the Magshimim "Research and POC" template. The POC was written in Python and actually run; the numbers are its output. What is real and what is simulated is stated explicitly.

Basic research: sequence of events

- **Prepare:** the paper is encrypted under a random key; the key is split into five shares (Shamir, threshold 3).
- **Distribute:** each centre receives an encrypted copy with a unique layout (watermark).
- **Open:** at the exam hour three custodians approve, the key is rebuilt, the paper opens.
- **Print:** in the digital scenario now; in the paper scenario, earlier.
- **Leak:** if a page leaks, its watermark is read and the centre identified.

Basic research: components

Paper Vault (encryption and key split), Custodian App (approval with HSM or smart card), Layout Marker (per-centre layout), Leak Reader (reads a leaked page).

Basic research: security challenges

- **Colluding custodians:** three together can open early; defence: different bodies and logged approvals.
- **Photo of a printed page:** the watermark must work here.
- **Retyping and answer keys:** many leaks spread as retyped text or "1-B, 2-D...". The watermark must survive these too; this is the question.

Five key features

Key split 3-of-5 over the prime field 2^127-1; encryption (SHA-256 stream in the POC, AES-256-GCM in deployment); layout watermark of 12 bits of centre id in trailing spaces; watermark reader; readability timeline (where and when a readable copy exists).

The POC: research question

Can early opening be prevented and the leaking centre identified, for the ways leaks really happen?

The POC: technologies and limits

Pure Python (`hashlib`, `secrets`). **Important limit:** paper, centres and leaks are synthetic. Shamir sharing is real; the cipher is a demo stream (AES-GCM in deployment); a "photo" is modelled as a layout-preserving copy.

The POC: results

Lab	Measured	Result
Split	2 / 3 / 5 custodians	2: stays closed · 3: opens · 5: opens; brute force ≈ 2.7×10^18 years
Photo	200 photo leaks, 900 centres	200 of 200 traced (100%)
Retype	the same 200 leaks retyped	0 traced
Retype	leak as an answer key	0 traced
Window	where a readable copy exists (illustrative timeline, not a measurement)	paper: at the press 72 h before, in trucks and strongrooms 36 h before · digital: 15 minutes before, on the centre server

The POC: conclusions

- ✅ **Claim 1, lock:** proven; no single custodian, and no two, can open.
- ❌ **Claim 2, trace:** fails where it matters. The watermark survives a photo, but leaks spread mostly as retyped text or answer keys, where it vanishes. A wording watermark would survive, but it changes questions between centres and harms fairness.
- ❌ **Claim 3, window:** fails. A paper exam must be printed hours before, so the lock protects the file and not the paper; the leak simply moves into the printing window.

Conclusion: ❌ falls at POC. The cryptographic core works, but two of three claims do not hold in the real world. **What would have saved it:** computer-based exams in controlled centres, where the key opens at the exam minute and no paper exists; then the lock is enough and the watermark is unnecessary.